
IT549: Deep Learning

Academic Year 2025-26 (Winter)

Instructor:	Dr. Arpit Rana (arpit_rana@daiict.ac.in)
Prerequisites:	Programming in Python, Machine Learning (<i>Not a Strict Criteria</i>)
Slot:	B.Tech. (ICT/ ICT with Minor in CS/ MnC) VI Semester, M.Tech. II Semester, and Ph.D. Students
Category:	ICT/ Technical Elective
Course Credits:	4 Credits (3-0-2)
Lectures (1 x 3):	TBD
Lab Session (1 x 2):	TBD
TA contact info:	TBD

Course Description:

Deep Learning aims to equip students with a comprehensive understanding of the principles, algorithms, and applications of neural networks and deep learning techniques. Through theoretical knowledge and practical experience, students will gain the expertise to effectively design, implement, and deploy neural network-based solutions across various domains. The course emphasizes practical application development through course projects and lab sessions, preparing students to understand and implement deep learning algorithms and methodologies.

Suggested Books:

- Deep Learning by Ian Goodfellow, Yoshua Bengio, Aaron Courville, The MIT Press [2016].
- Deep Learning with Python by François Chollet, Manning Publications, 2nd Edition [2021].
- Deep Learning from Scratch by Seth Weidman, O'Reilly, 1st Edition [2019]

Course Outcomes:

- Understand the advantages of deep learning.
- Gain in-depth knowledge of advanced architectures, including CNNs, RNNs, LSTMs, GRUs, Attention mechanisms, Transformers, and Autoencoders (at a high level).
- Theoretical and practical understanding of the architectures and insights into design choices for better model development.

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- Learn essential model training concepts, including regularization, dropout, data augmentation, batch normalization, and hyperparameter tuning, which are explored for effective optimization.
- Hands-on demos using TensorFlow/ PyTorch on images, text, time series, language data, etc., are included for all architectures, equipping students with practical skills to excel in deep learning.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X		X				X			X

Evaluation Scheme (Tentative):

Ability to Understand and Apply Theoretical Concepts		Ability to Implement and Solve Real Problems	
In-Sem Exams (25 %)	Endterm Exam (30 %)	Challenges (30 %)	Project Viva (15%)

Tentative Detailed Course Plan¹:

Units	Topics	# Lectures
Module 1: Foundations of Deep Learning and Neural Computation	Algebraic, Probabilistic, and Graphical View of Data. Importance of deep learning, representation learning from large datasets, case studies & modern applications; artificial neurons, inputs, weights, and activation functions; relationship with linear models; multilayer neural networks and matrix–vector operations; input–hidden–output layers; motivation for deeper architectures.	6
Module 2: Training Neural Networks: Backpropagation & Optimisation	Hyperparameters vs. parameters; supervised training pipeline; loss functions; detailed mathematics of backpropagation; computational graph perspective; training challenges; vanishing/exploding gradients; batch normalisation; improved optimisers (SGD variants, momentum, Adam); learning-rate schedules; initialisation strategies (Xavier/ He).	6

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Module 3: Regularisation, Generalisation & Practical Training Strategies	Overfitting and underfitting; reducing model size; L1/L2 regularisation; dropout; early stopping; data augmentation; transfer learning; unsupervised pre-training; pre-training auxiliary tasks; fine-tuning strategies for modern architectures.	6
Module 4: Convolutional Neural Networks (CNNs)	Motivation; convolution operation; images as rank-3 tensors; filters/kernels; strides, padding, dilation; feature maps and receptive fields; stacking layers; pooling layers and their varieties; modern CNN architectures (LeNet, AlexNet, VGG, ResNet: conceptual, not detailed blueprints); training CNNs, hyperparameters, memory constraints; overfitting in vision; data augmentation; pre-trained CNNs; transfer learning in vision tasks.	6
Module 5: Recurrent Models for Sequences — RNN, LSTM, GRU	Recurrent neurons and layers; unfolding through time; training RNNs; time-series forecasting with simple and deep RNNs; vanishing/exploding gradients in sequences; LSTM and GRU architectures; forecasting and sequence modeling examples.	6
Module 6: Sequence-to-Se quence Models & Attention Mechanisms	Encoder-decoder networks; bidirectional RNNs; teacher forcing & scheduled sampling; beam search; attention mechanism: intuition and mathematics; self-attention; comparison of recurrent vs. attention-based architectures; positional encodings.	6
Module 7: Transformer Architecture and Modern Deep Learning Paradigms	The Transformer architecture, multi-head attention, feed-forward blocks, residual connections, training considerations (batching, masking, optimisers), pre-training and fine-tuning large models, and a conceptual overview of Vision Transformers (ViT) and GPT-style models.	6

1 — This course plan is subject to change without notice.

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Preliminary Schedule of the Course²:

Week	Lecture	Lab
Week-1 [5 Jan 2026]	Course Admin Policy. Algebraic, Probabilistic, and Graphical View of Data.	– No lab –
Week-2 [12 Jan 2026]	Foundations of Deep Learning and Neural Computation I	– No lab – (Group formation and domain finalization through Google Form)
Week-3 [19 Jan 2026]	Foundations of Deep Learning and Neural Computation II	Linguistic Preprocessing, Text Vectorization, and Embeddings
Week-4 [26 Jan 2026]	Training Neural Networks: Backpropagation & Optimisation I	Pre-training and Fine-tuning
Week-5 [2 Feb 2026]	Training Neural Networks: Backpropagation & Optimisation II	– Buffer Lab –
Week-6 [9 Feb 2026]	First In-Semester Examination (9 th Feb - Friday to 12 th Feb - Tuesday)	
Week-6 [16 Feb 2026]	Regularisation, Generalisation & Practical Training Strategies	100-Minute ML Development Challenge
Week-7 [23 Feb 2026]	Convolutional Neural Networks (CNNs) – I	Image Vectorization or Image Tensors: Using Pre-trained CNNs, Fine-tuning, and More
Week-8 [2 Mar 2026]	In-semester Break (Entire week)	
Week-9 [9 Mar 2026]	Convolutional Neural Networks (CNNs)-- II	Object Detection and Image Segmentation
Week-10 [16 Mar 2026]	Second In-Semester Examination (18 th Mar Wednesday to 23 rd Mar Monday)	

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Week-11 [23 Mar 2026]	Recurrent Models for Sequences: RNN	100-Minute ML Development Challenge
Week-12 [30 Mar 2026]	Recurrent Models for Sequences: LSTM, GRU	Working with Sequences or Time-Series Data (Course Project Announcement)
Week-13 [6 Apr 2026]	Sequence-to-Sequence Models & Attention Mechanisms.	Unsupervised Learning using Encoder-Decoder: An application for demonstration
Week-14 [13 Apr 2026]	Transformer Architecture and Modern Deep Learning Paradigms – I	Training Transformers
Week-15 [20 Apr 2026]	Transformer Architecture and Modern Deep Learning Paradigms – II	100-Minute ML Development Challenge
Week-16 [27 Apr 2024]	Last date of Classes, Labs, and Tutorials (29th Apr 2026)	
End-Semester Examination (30th Apr 2026 to 8th May 2026)		

2 — This weekly course plan is subject to change without notice.

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Challenges and Course Project Viva [20% + 25% = 45%]

The following points regarding the course project are worth noting.

Student Groups (Just for Challenges)

- Students will be asked to form groups of two members. Each group will be involved in 100-minute ML development challenges.
 - 45% of your marks will be based on team efforts, so choose your members wisely.
 - Teams will remain unchanged throughout the semester once registered. No requests for changes will be entertained.
- Team registration will open the second week after classes start, and you must register your team via a Google form within three days of the announcement.
- During lab hours, there will be three machine-learning challenges. The three most proficient teams shall be acknowledged with a bonus of up to 3% on their respective scores.

Course Project [25%]

- There is only one course project: an End-to-End ML application (full-stack, i.e., front-end and Back-end, build and deploy) that solves a data-driven problem with high social impact.
- Each group must select a thematic domain: Finance, E-commerce, Healthcare, Pharma, Sports, Entertainment, Renewable Energy, Oil & Gas, Automobile, Agriculture, FMCG, Cybersecurity, Social Media, Supply Chain, Manufacturing, Business Automation, or any other exciting and valuable socially aligned domain.
- Each group will define the problem in their selected domain and collect dataset(s) from reliable sources, including publicly available ones (no two groups can work on the same dataset). You are encouraged to gather additional data to enhance your dataset and better address the problem.
- Each group must add a novel contribution to their project and compare their work with the existing baselines.
- The project guideline document will provide information on domain allocation, general instructions, evaluation criteria, and other protocols.

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Submission

- Each group will submit the project code repository, along with a comprehensive README file, on GitHub Classroom. Submission instructions will be provided in the project guideline document.
- Evaluation will primarily be a Chalk Talk with the Instructor. Any student may be asked questions about any aspect of the project.
- Late submissions (up to 24 hours) will incur a 20% penalty.